

Project Initialization and Planning Phase

Date	01 JUNE 2026
Team ID	XXXXXX
Project Name	<i>OptiCrop: Smart Agricultural Production Optimization Engine</i>
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement):

I am	I'm trying to	But	Because	Which makes me feel
a farmer who wants to choose the most suitable crop for cultivation based on soil and climate conditions	identify the best crop using soil nutrients (N, P, K) and environmental factors like temperature, humidity, pH, and rainfall	I lack a reliable system that can analyze these parameters and give accurate crop recommendations	there is no easy to use intelligent system that applies machine learning to agricultural data for real time crop prediction	uncertain and dependent on guesswork when selecting crops, which may lead to low yield and resource loss

Project: **OptiCrop: Smart Agricultural Production Optimization Engine**

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a farmer who wants to grow the most suitable crop for my field	select the best crop based on soil nutrients (N, P, K) and environmental conditions	traditional decision-making is based on experience and guesswork, which may not consider all important factors together	there is no automated system that analyzes soil and climate data to recommend the most suitable crop accurately	uncertain and worried about choosing the wrong crop and facing low yield or financial loss
PS-2	an agricultural advisor who guides farmers on crop selection	provide accurate and data-driven crop recommendations quickly	analyzing soil tests, weather conditions, and seasonal factors manually is time-consuming and complex	there is no intelligent system that integrates multiple agricultural parameters and predicts the best crop	overwhelmed with data and worried about giving inaccurate advice
PS-3	a farm owner who wants to maximize yield and make better use of resources	increase crop productivity and profitability by choosing the right crop for my field conditions	wrong crop selection leads to poor yield, inefficient use of water, fertilizers, and other resources	there is no data-driven recommendation tool that considers soil, climate, and seasonal patterns together	frustrated and concerned about low productivity and wasted resources